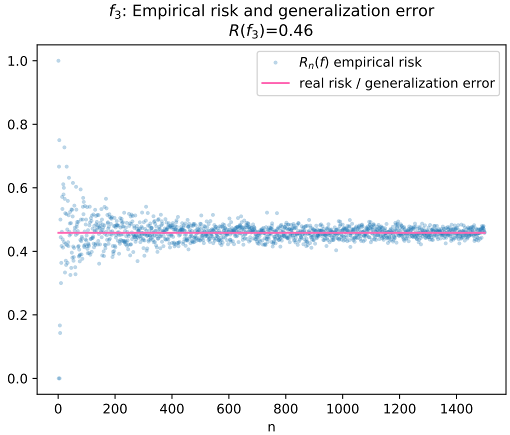
Fondamentaux théoriques du machine learning



Overview of lecture 3

- Recap on risks and Bayes estimators
    - Binary classification example
    - Problem of empirical risk minimization
    - Bayes estimators in two specific cases
        - Classification with "0-1" loss
        - Regression with squared loss
    - Statistical analysis of OLS

Supervised learning

- The dataset D_n is a collection of n samples $\{(x_i, y_i), 1 \leq i \leq n\}$, that are assumed **independent and identically distributed** draws of from the **joint random variable** (X, Y).
- the law of (X, Y) is unknown, we can note it ρ.

Risks (reminder of the definition)

Let l be a loss function.

- The **empirical risk (ER)** of an estimator f writes

$$R_n(f) = \frac{1}{n}\sum_{i=1}^{n} l(y_i, f(x_i)) \tag{1}$$

- The **risk** (or **statistical risk, generalization error, test error)** of estimator f writes

$$R(f) = E_{(X,Y)\sim\rho}[l(Y, f(X))] \tag{2}$$

Generalization of the penalty shootout example

We consider the following random variable (X, Y).

- $\mathcal{X} = \{0, 1\}$, $\mathcal{Y} = \{0, 1\}$.
- $X \sim B(\frac{1}{2})$,

$$Y = \begin{cases} B(p) \text{ if } X = 1 \\ B(q) \text{ if } X = 0 \end{cases}$$

With $B(p)$ a Bernoulli law with parameter p.

We consider 3 **estimators** :

- $$f_1 = \begin{cases} 1 \text{ if } x = 1 \\ 0 \text{ if } x = 0 \end{cases}$$
- $$f_2 = \begin{cases} 0 \text{ if } x = 1 \\ 1 \text{ if } x = 0 \end{cases}$$
- $$\forall x \in \mathcal{X}, f_3(x) = 1 \tag{3}$$

Exercice 1 :

We observe the following dataset :

$$D_4 = \{(0,1),(0,0),(0,0),(1,0)\}$$

Compute the **empirical risks** $R_4(f_1)$, $R_4(f_2)$, $R_4(f_3)$ with the "0-1" loss.

$$R_n(f) = \frac{1}{n}\sum_{i=1}^{n} l(y_i, f(x_i))$$

Law of total probability

If for instance $\Omega = A \cup B \cup C$ and A, B, C are mutually exclusive and collectively exhaustive (système complet d'événements), then

$$P(X) = P(X \cap A) + P(X \cap B) + P(X \cap C) \tag{4}$$

`https://en.wikipedia.org/wiki/Law_of_total_probability`
Ω is the **sample space**.

Conditional probabilities

$$P(A \cap B) = P(A|B)P(B) \tag{5}$$

Real risks

Exercice 2 : Now, compute the real risks $R(f_1), R(f_2), R(f_3)$.

$$R(f) = E[l(Y, f(X))] \tag{6}$$

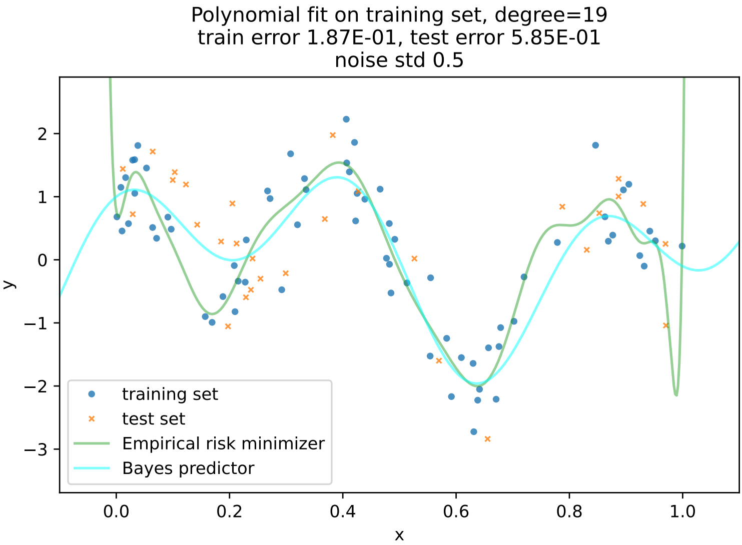
$$
\begin{aligned}
R(f_1) &= E[l(Y, f(X))] \\
&= 1 \times P\Big(l(Y, f(X)) = 1\Big) + 0 \times P\Big(l(Y, f(X)) = 0\Big) \\
&= 1 \times P(Y \neq f(X)) + 0 \times P(Y = f(X)) \\
&= P(Y \neq f(X))
\end{aligned} \tag{7}
$$

Random variables or deterministic quantities

- $R_4(f)$ (empirical risk) **depends** on D_4. If we sample another dataset, $R_4(f)$ is likely to change, it is a **random variable**.
- $R(f)$ (generalization error) is **deterministic**, given the joint law of (X, Y).

Optimization problem

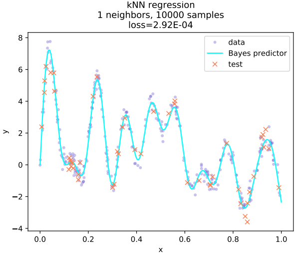
- The smaller the generalization error $R(f)$ is, the better f is.
- The situation is more tricky for $R_n(f)$: it is not obvious that as estimator that has a very small empirical risk $R_n(f)$ has a small generalization error $R(f)$! This is the problem of **overfitting**.


Polynomial fit on training set, degree=19
train error 1.87E-01, test error 5.85E-01
noise std 0.5
y
x
2
1
0
−1
−2
−3
0.0
0.2
0.4
0.6
0.8
1.0
training set
test set
Empirical risk minimizer
Bayes predictor

Empirical risk minimization

Look for f_n that minimizes $R_n(f)$.

Not all function approximations are based on finite datasets consists in empirical risk minimization ! (nearest neighbors are not)



Optimization problem

Empirical risk minimization (ERM) : finding the estimator f_n that minimizes the empirical risk R_n.
This raises important questions :

- 1) does f_n have a good generalization error $R(f_n)$?
- 2) how can we have guarantees on the generalization error $R(f_n)$?
- 3) how can we find the empirical risk minimizer f_n ?
- 4) is it even interesting to strictly minimize R_n ?

Generalization error

Question 1) Does f_n have a good generalization error $R(f_n)$?
This will depend on :

- the number of samples n
- the shape of f (the map such that $Y = f(X)$), in particular on its **regularity** (or equivalently the distribution ρ)
- the dimensions of the input space and of the output space.
- the space of functions where f_n is taken from.

Statistical bounds

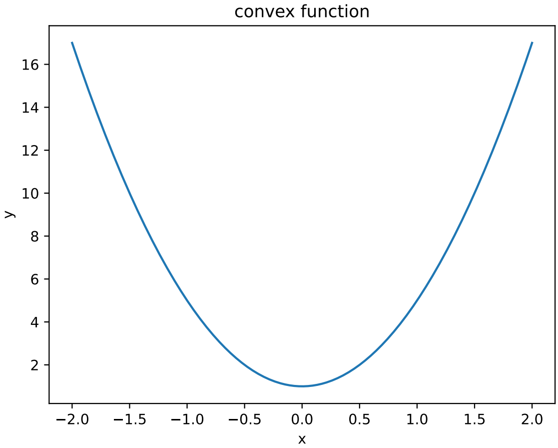
Questions 2) How can we have guarantees on the generalization error $R(f_n)$?
By making **assumptions** on the problem (learning is impossible without making assumptions), for instance assumptions on ρ.

Optimization

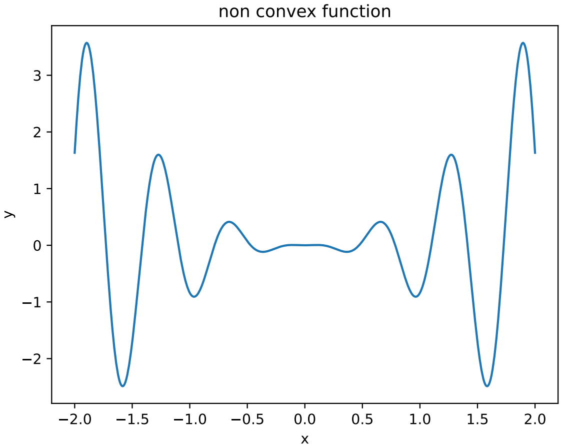
Question 3) how can we find the empirical risk minimizer f_n ?
By using an optimization algorithm or by solving the minimization in closed-form.

Convex functions

Convex functions are easier to minimize.



Non convex functions



What is convex here ?

In this context, the convexity that is involved is the dependence of R_n in g. More precisely, for instance if g depends on $\theta \in \mathbb{R}^d$, e.g. $g(x) = \langle \theta, x \rangle$, the convexity is that of

$$\theta \mapsto R_n(\theta) \tag{8}$$

Example (ordinary least squares) :

$$R_n(\theta) = \frac{1}{n} \sum_{i=1}^{n} (\langle \theta, x_i \rangle - y_i)^2 \tag{9}$$

with $x_i \in \mathbb{R}^d$, $y_i \in \mathbb{R}$.

Optimization error

Question 4) is it even interesting to strictly minimize R_n ?
Most of the time it is **not**, as we are interested in R, not in R_n, so we should not try to go to machine precision in the minimization of a quantity that is itsself an approximation !
(This is linked to the **estimation error** that is often of order $\mathcal{O}(1/\sqrt{n})$.)

Bayes risk

We define the **Bayes estimator** f^* by

$$f^* \in \underset{f : X \to Y}{\arg\min}\, R(f)$$

with $f : X \to Y$ set of measurable functions. The **Bayes risk** is $R(f^*)$.
Fundamental problem of supervised learning : Approximate f^* given only D_n.

Bayes estimators

As we have admitted during the TPs :

- if we know the law ρ of (X, Y)
- if the loss l is well known (e.g. the squared loss, the "0-1" loss)

Then we can sometimes explicitely derive en expression of the Bayes estimator, as in the first two practical sessions.

Practical sessions

During the practical sessions with experimented with several notions related to risks in supervised learning.

- **TP1 :** given a problem, find the Bayes estimator
- **TP2 :** given a problem, compare some estimator (OLS, Ridge) to the Bayes estimator.

In both cases, we assumed that we had a **perfect statistical knowledge** of the problems.

Practical situations

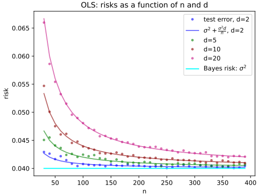
Hence, if we knew ρ in a situation as just described, **learning would not be necessary**.

But in concrete problems, we do not now ρ. **Why even mention Bayes estimators and Bayes risks then ?**

Because in some contexts we can have a good idea of whether we can have a satisfactory approximation of f^* based on the dataset only, aka whether **learning is possible**.

Example : for OLS, we have experimentally seen that (in the specific statistical hypothesis that we assumed, named the "fixed design") :

$$E[R(\hat{\theta})] - R(\theta^*) = \frac{\sigma^2 d}{n} \tag{10}$$



So for instance :

- if $d << n$, then yes, learning is possible with OLS
- if $d = n$, then the OLS is only half as good as f^* (because $R(\theta^*) = \sigma^2$).

Bayes estimators

We will compute the Bayes estimator for :

- the squared-loss
- the "0-1" loss

We assume again that $(X, Y) \sim \rho$. We look for the predictor f^* that mininimizes :

$$R(f) = E_{(X,Y)\sim\rho}[l(Y, f(X)] \tag{11}$$

Law of total expectation

https://en.wikipedia.org/wiki/Law_of_total_expectation

Law of total expectation

$$
\begin{aligned}
R(f) &= E_{X,Y}[l(Y, f(X))] \\
&= E_X\Big[E_Y[l(Y, f(X))|X]\Big] \\
&= E_X\Big[h_f(X)\Big]
\end{aligned}
\tag{12}
$$

$h_f(X) = E_Y[l(Y, f(X))|X]$ is a function of X, that depends on f.

We might minimize h independently for all values x of X !

$$
f^*(x) = \underset{z \in \mathcal{Y}}{\arg\min}\, E_Y\big[l(Y, z)|X = x\big] \tag{13}
$$

Classification with "0-1" loss

$$f^*(x) = \underset{z \in \mathcal{Y}}{\arg\min}\, E_Y \left[l(Y, z) | X = x \right] \tag{14}$$

We assume that $Y \in \mathcal{Y} \in \mathbb{N}$ and that l is the "0-1" loss.

Exercice 3 : What is $f^*(x)$?

Regression with squared loss

$$f^*(x) = \underset{z \in \mathcal{Y}}{\arg\min}\, E_Y\left[l(Y, z)|X = x\right] \tag{15}$$

We assume that $Y \in \mathcal{Y} \in \mathbb{R}$ and that l is the squared loss.

Exercice 4 : What is $f^*(x)$?

OLS

- $\mathcal{X} = \mathbb{R}^d$
- $\mathcal{Y} = \mathbb{R}$.
- $l(y, y') = (y - y')^2$ (squared loss)
- $$F = \{x \mapsto \theta^T x, \theta \in \mathbb{R}^d\}$$

OLS

The dataset is stored in the **design matrix** $X \in \mathbb{R}^{n \times d}$.

$$X = \begin{pmatrix} x_1^T \\ \dots \\ x_i^T \\ \dots \\ x_n^T \end{pmatrix} = \begin{pmatrix} x_{11}, \dots, x_{1j}, \dots x_{1d} \\ \dots \\ x_{i1}, \dots, x_{ij}, \dots x_{id} \\ \dots \\ x_{n1}, \dots, x_{nj}, \dots x_{nd} \end{pmatrix}$$

The vector of predictions of the estimator writes $y_{pred} = X\theta$. Hence,

$$\begin{aligned} R_n(\theta) &= \frac{1}{n} \sum_{i=1}^{n} (y_i - \theta^T x_i)^2 \\ &= \frac{1}{n} ||y - X\theta||_2^2 \end{aligned}$$

OLS estimator

The objective function $R_n(\theta)$ is convex in θ.

Proposition

Closed form solution
*We X is injective, there exists a unique minimiser of $R_n(\theta)$, called the **OLS estimator**, given by*

$$\hat{\theta} = (X^T X)^{-1} X^T y \tag{16}$$

Statistical setting : fixed design, linear model (TP2)

Assumptions :

- **Linear model** : $\exists \theta^* \in \mathbb{R}^d$,

$$y_i = {\theta^*}^T x_i + \epsilon_i, \forall i \in [1, n]$$

  and ϵ_i is a centered noise (or error) ($E[\epsilon_i] = 0$) with variance σ^2.
- Fixed design X (or random design where X is also random)

In this setting, we can now derive :

- 1) the Bayes predictor
- 2) the expected value of the OLS estimator
- 3) its excess risk (difference of its risk with Bayes risk)

1) Bayes predictor

With the squared loss, we always have that the Bayes predictor is the conditional expectation (also in lecture_.pdf section 3.1.3.)

$$f^*(u) = E[Y|x = u] \tag{17}$$

1) Bayes predictor

$$\begin{aligned} f^*(u) &= E[Y|x = u] \\ &= E[x^T\theta^* + \epsilon|x = u] \\ &= E[x^T\theta^*|x = u] + E[\epsilon|x = u] \\ &= u^T\theta^* \end{aligned} \tag{18}$$

1) Bayes risk

Fixed design risk : the inputs are fixed (it is also possible to use a random design).

$$
\begin{aligned}
R^* &= E_y[(y - f^*(X))^2] \\
&= E_\epsilon[(X^T\theta^* + \epsilon - X^T\theta^*)^2] \\
&= E_\epsilon[\epsilon^2] \\
&= \sigma^2
\end{aligned}
\tag{19}
$$

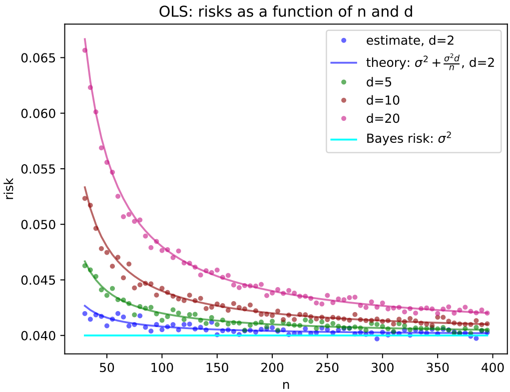
2) Expected value of $\hat{\theta}$

$$
\begin{aligned}
E[\hat{\theta}] &= E[(X^TX)^{-1}X^Ty] \\
&= E[(X^TX)^{-1}X^T(X\theta^* + \epsilon)] \\
&= E[(X^TX)^{-1}X^T(X\theta^*)] + E[(X^TX)^{-1}X^T\epsilon] \\
&= E[(X^TX)^{-1}(X^TX)\theta^*] + (X^TX)^{-1}X^TE[\epsilon] \\
&= E[\theta^*] \\
&= \theta^*
\end{aligned}
\tag{20}
$$

We conclude that the OLS estimator is an **unbiased estimator** of θ^*.

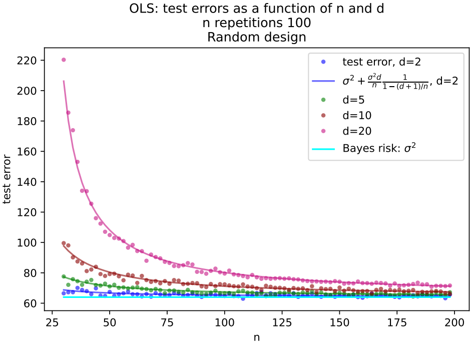
3) Excess risk (fixed design)

$$E\left[R(\hat{\theta})\right] - R(\theta^*) = \frac{\sigma^2 d}{n} \tag{21}$$



3 bis) Excess risk (random design)

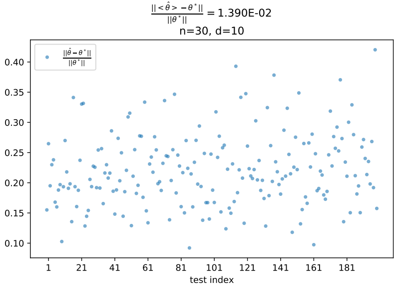
$$E\left[R(\hat{\theta})\right] - R(\theta^*) = \frac{\sigma^2 d}{n} \frac{1}{1 - (d+1)/n} \tag{22}$$



4) Variance

$$Var(\hat{\theta}) = \frac{\sigma^2}{n}\Sigma^{-1} \tag{23}$$

with $\Sigma = \frac{1}{n}X^TX \in \mathbb{R}^{d\times d}$.



Issues in high dimension

The problem can become **ill-conditioned**.
When d is large (for instance when $\frac{d}{n}$ is close to 1), then

- the amount of excess risk is not way smaller than σ^2.
- if $d = n$ and X^TX is invertible, we can fit the training data exactly, which is bad for generalization.

If $d > n$, X^TX is not invertible, we do not have a unique closed form solution anymore, we can might have a **subspace** of solutions.
Remark : with $d << n$, it is also possible to have an ill-conditioned matrix (for instance if X has colinear columns).